LAFUSION 2026 · FOURTH LATIN AMERICAN WORKSHOP ON INFORMATION FUSION

From Baseline to HIT-Tracker

An Experimentally Optimized Pipeline for Infrared Multi-UAV Tracking

Ighor de Souza Ribeiro[1] Rigel Procópio Fernandes[2] Gabriel de Sapienza Luna[1,3]

[1]CIAA [2]DSAM [3]UFRJ

follow the slides

FROM ONE UAV TO A SWARM



SIGE 2026
one UAV · visible spectrum



This work
UAV swarms · infrared

From one target to many identities

THE PROBLEM

Small · textureless · look-alike drones

30	22,839	478,548
held-out sequences	frames	ground-truth boxes

Goal: keep the identity of every drone

MOTA AND IDF1: WHAT WE ATTACK

$$\mathrm{MOTA} = 1 - \frac{\mathbf{FN} + \mathbf{FP} + \mathbf{IDs}}{\mathrm{GT}}$$

all errors (official ranking)

$$\mathrm{IDF1} = \frac{2\,\mathrm{IDTP}}{2\,\mathrm{IDTP} + \mathrm{IDFP} + \mathrm{IDFN}}$$

identity kept over time

FN, FP detector: **H**, **T**

IDs, IDF1 association: **I**

Detector fixes FN and FP; association fixes identity switches

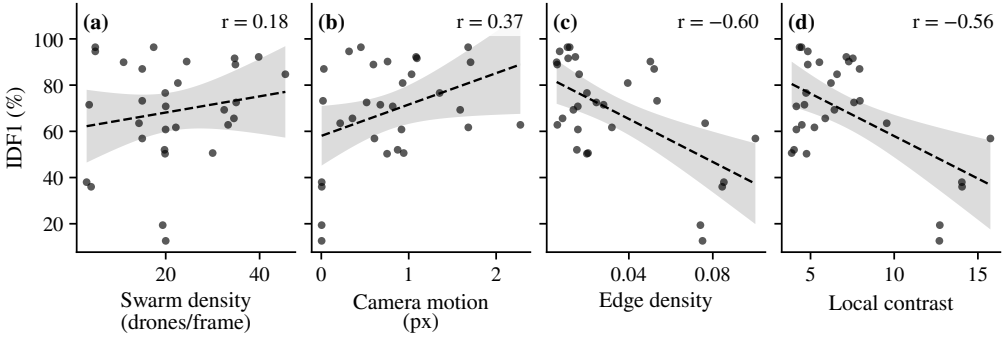
BASELINE: YOLO26N + TRACKERS

YOLO26n detector → **Online tracker** six candidates → **Tracks** boxes + IDs

Tracker	MOTA (%)	IDF1 (%)	IDs	FPS
BoT-SORT	**76.0**	**71.2**	2355	45.05
FastTracker	72.7	50.5	3948	51.96
ByteTrack	71.7	49.4	4582	**52.74**
OC-SORT	71.5	51.8	4454	49.61
Deep OC-SORT	71.2	52.5	4165	48.70
TrackTrack	63.1	68.9	**836**	43.54

BoT-SORT: best MOTA and IDF1 → our baseline

ROOT CAUSE



Identity loss follows background clutter, not swarm density

STRATEGY: H, T AND I

detector

H

Hard-example mining

cheap fine-tuning

detector

T

Tiled inference

slower inference

association

I

Improved association

no retraining,
negligible cost

Fight clutter in the detector; fix identities in the association

H: HARD-EXAMPLE MINING



White boxes: ground-truth drones. Training sequences used for hard-example mining (HEM).

rank training sequences by **MOTA**

↓

3,846 hard images
+ **5,000** random

↓

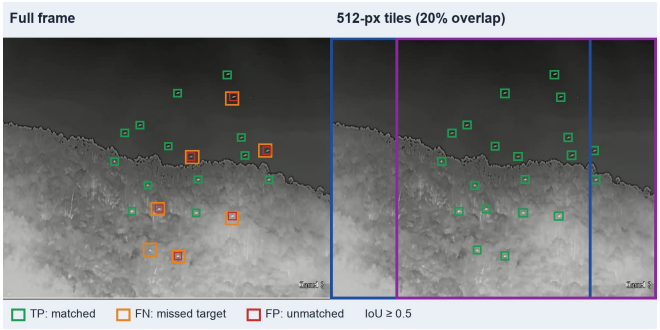
fine-tune **8 epochs**
on one T4 GPU

65,554 → 61,597
false negatives

76.0 → 76.7
MOTA (%)

Cheap retraining: fewer misses

T: TILED INFERENCE (SAHI)



Tile	MOTA	IDF1	IDs
320	76.4	77.5	864
384	77.6	79.5	642
448	77.9	**80.8**	**592**
512	**79.6**	80.3	712
576	**79.6**	79.9	718

Small drones at native resolution: MOTA 76.7 → 79.6

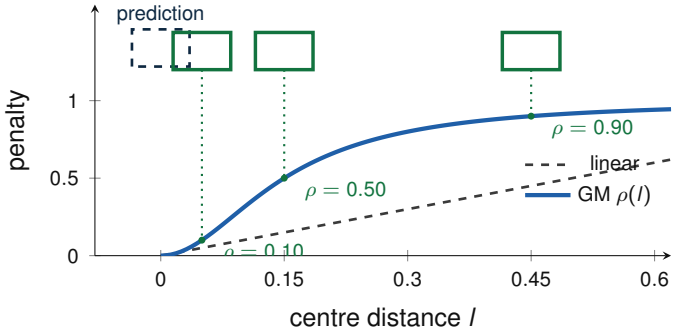
I: ROBUST ASSOCIATION COST

$$C = (1 - \gamma)\, C_{\mathrm{IoU}} + \gamma\, \rho(l)$$

$$\rho(l) = \frac{l^2}{l^2 + \sigma^2}$$

$\sigma = 0.15$ $\gamma = 0.55$

l: distance between predicted and detected centres



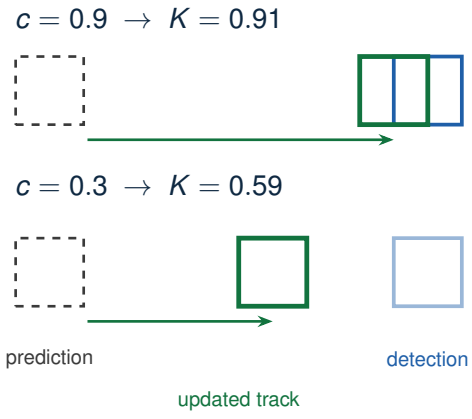
Near detections are almost free; far ones are capped at 1

I: NSA-KALMAN

$$R' = \max(1 - c,\ 0.05)\, R$$

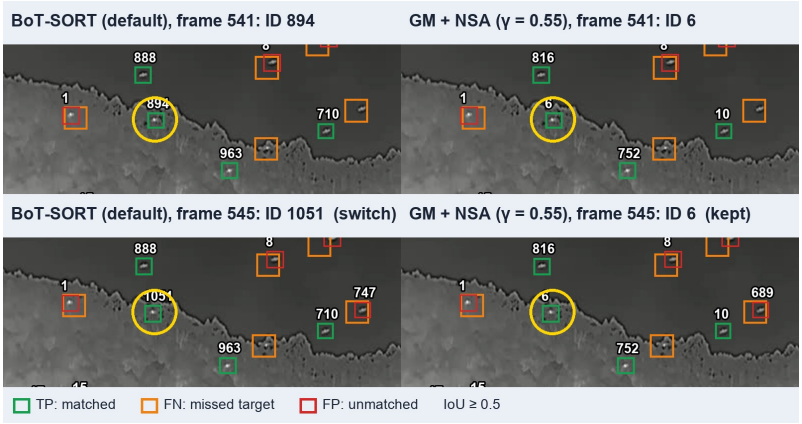
$$K = PH^\top \left(HPH^\top + R'\right)^{-1}$$

high $c \rightarrow$ small $R' \rightarrow$ large K



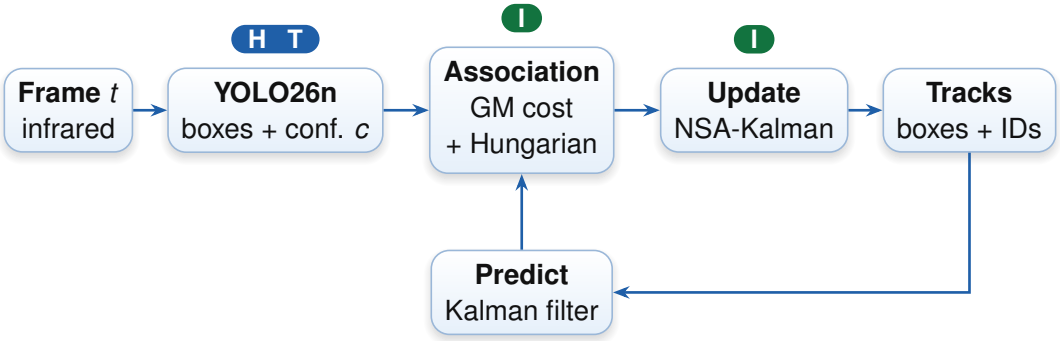
Confident detections pull the track more; weak ones pull it less

I IN ACTION



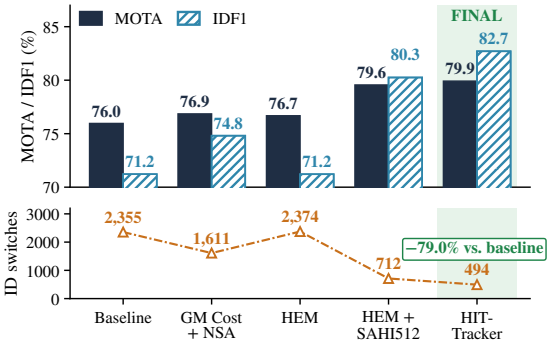
IDF1 71.2 → 74.8 and IDs 2,355 → 1,611, without retraining

THE HIT-TRACKER PIPELINE



H and T in the detector, I in the association

HIT-TRACKER RESULTS



+3.9 p.p.
MOTA: 76.0 → 79.9

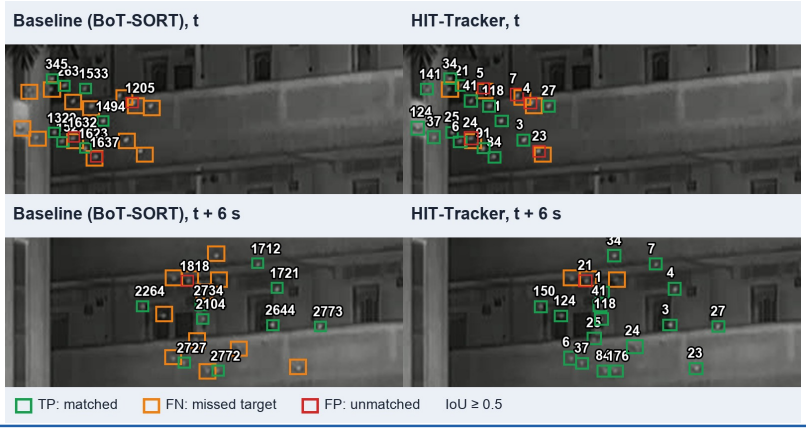
+11.5 p.p.
IDF1: 71.2 → 82.7

−79.0%
IDs: 2,355 → 494

45.05 → 14.48 FPS (tiles)

Better in 27 of 30 sequences (MOTA) and 29 of 30 (IDs)

SEEING THE DIFFERENCE



Baseline IDs keep changing on the facade; HIT-Tracker keeps them

LIMITATIONS AND FUTURE WORK

Limitations

- Internal held-out split (no test server)
- Settings chosen on the same sequences: optimism < 0.7 p.p.
- One dataset, one training run
- FPS from one laptop

Future work

- Larger YOLO26 variants
- Retraining with CLAHE and gamma
- GM cost in end-to-end Transformer trackers
- New infrared benchmarks and embedded hardware

Next: independent benchmarks and embedded deployment

CONCLUSIONS

1. **Clutter**, not density, breaks identities
2. **Detection** (H + T) brings most of the gain
3. **Association** (I) adds identity at negligible cost

Largest identity gains in the most cluttered sequences



Thank you!

Questions?

Ighor de Souza Ribeiro

ighoribeiropb@gmail.com

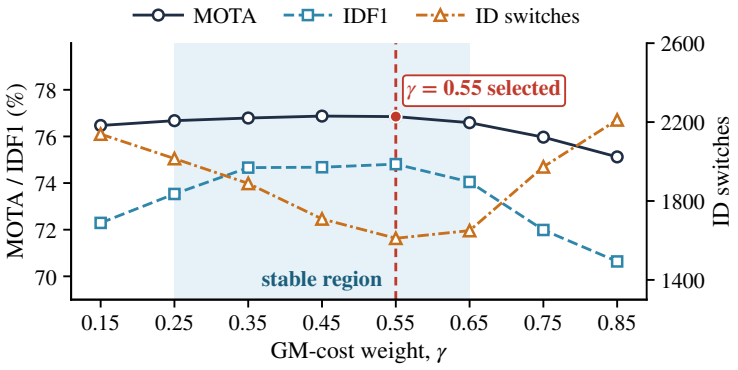
Code and configurations
github.com/IghorRibeiro/
DroneTrackingMOT

BACKUP: WHAT DID NOT WORK

Configuration	MOTA (%)	IDF1 (%)	IDs
Baseline	**76.0**	**71.2**	**2355**
MOG2	−168.6	16.0	5048
EMA	2.4	38.0	4882
Dense optical flow	22.6	25.8	5544
GM + NSA (reference)	76.9	74.8	1611
Static track filter	65.7	68.9	1816
Lost-query bank	76.0	72.4	3464
Adaptive threshold	**77.0**	**74.9**	**1535**

Detection-only (3,846 hard images): mAP50 0.547 $\rightarrow$ 0.529 (CLAHE, gamma, unsharp) and 0.167 (sharpen)

BACKUP: γ SWEEP



MOTA stable for γ from 0.25 to 0.65; fewest IDs at 0.55

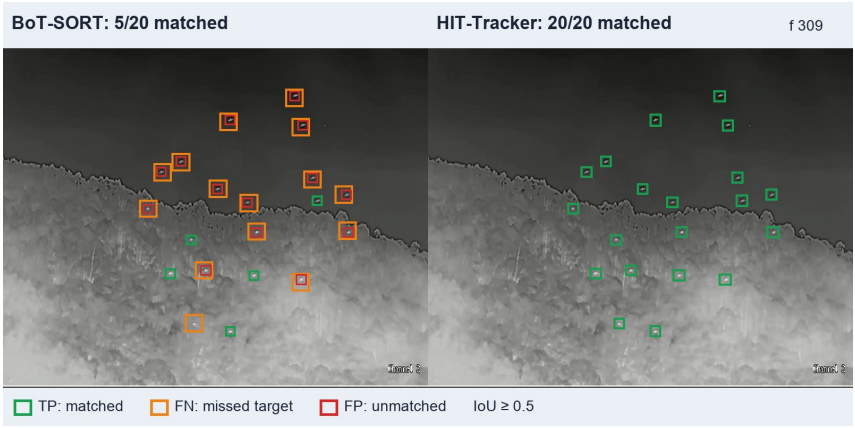
BACKUP: ROBUSTNESS AND STATISTICS

- **Split-half analysis** (1,000 random 15/15 splits): optimism $+0.1$ MOTA / $+0.6$ IDF1 for γ; $+0.3$ / $+0.3$ for the tile size. $\gamma = 0.45$ and 576-px tiles lie within 0.1 p.p. in MOTA.
- **GM + NSA vs. baseline**: IDF1 $p = 0.004$, IDs $p = 0.026$, MOTA $p = 0.49$ (Wilcoxon).
- **HEM vs. baseline**: $p = 0.25$. **HIT-Tracker vs. HEM + SAHI512**: IDF1 $p = 0.040$, IDs $p = 0.011$, MOTA $p = 0.045$.
- **Clutter tertiles**: ΔIDF1 $+10.8$, $+9.3$, $+16.7$ p.p. (low to high edge density).
- **Limitations**: internal held-out split; one dataset; one training run; one laptop.

BACKUP: LITERATURE

Method	MOTA (%)	Note
PP-Tracker	67.9	Reported by authors
Strong Baseline	76.1	Reported by authors
Dist-Tracker	81.3	2nd place, official test
Context-Aware ID	**84.9**	1st place, official test

BACKUP: UAV240, FRAME 309



Baseline: 5 of 20 drones matched; HIT-Tracker: all 20